

Model Validation Report: LLM Extraction of Credit Risk Inputs

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1. Model Overview

This report tests whether an LLM can extract the six inputs of the Altman Z'' -score from 10-K filings accurately enough for credit screening, and whether Z'' itself predicts bankruptcy. The Z'' -score combines four ratios from the balance sheet and income statement into one number. Z'' -scores above 2.60 fall in the safe zone, scores from 1.10 to 2.60 in the grey zone, and scores below 1.10 in the distress zone. These are the cutoffs commonly applied to Z'' without its constant term (Section 4). Z'' suits this study for three reasons: (1) all of its inputs are printed in the filing, so each can be checked against the company's XBRL data, while the original Z-score needs the stock's market value; (2) it drops the sales-to-assets ratio, which Altman found sensitive to industry differences such as retail versus manufacturing, so it suits a sample of retail and software firms; (3) it is public and widely used, so it serves as a reproducible stand-in for a bank's internal credit model.

With the correct statements as input, both LLMs matched the filed values for 99.9–100% of fields and placed no firm in the wrong risk zone. However, retrieval was the weak point. In the first run, the pipeline failed to find the balance sheet and income statement in some filings. For those firms, the model received text that did not contain the needed numbers, so it returned no value. Twelve of 148 firms could not be scored. After the search was fixed (Section 5), every firm was scored. The extraction results apply to clean 10-K filings from firms whose statements show all six inputs. In the backtest, Z'' ranked firms by their risk of filing for bankruptcy in the following year. The AUC (area under the ROC curve) measures this ranking. It is the chance that a randomly chosen firm that went bankrupt scores as riskier than a randomly chosen firm that did not, so 0.5 means chance and 1.0 means a perfect ranking. Z'' had an AUC of 0.82 (95% interval 0.75–0.88) on FY2023 values and 0.76 (0.60–0.89) on FY2024 values. Refitting the score's weights on FY2023 data raised the FY2024 AUC to 0.91 (0.81–0.96).

The system under review extracts the six Z'' inputs from a company's Form 10-K for use in credit risk screening. It has four stages: (1) download the 10-K from the SEC, (2) find the balance sheet and income statement in the filing and cut them out, (3) send that text to

an LLM, which returns the six values, and (4) compute Z'' and its zone from those values. The extraction tests in Section 5 cover stages 2 through 4, and the backtest in Section 6 tests Z'' itself.

Three methods were compared at stage 3:

Method	Provider	Access
GPT-6 Luna	OpenAI	OpenRouter API
Claude Haiku 4.5	Anthropic	OpenRouter API
Rule-based challenger	Written for this study	Local code, no API

The validation does not cover credit decisions made without human review, foreign issuers that file Form 20-F, or financial firms (SIC 6000–6999), since Altman (2018) describes Z'' as a model for manufacturing and non-manufacturing industrial firms.

A bank needs this kind of extraction where no XBRL exists, as with the financial statements private borrowers submit with a loan application. Public 10-K filings stand in for that setting here because their XBRL data supply an answer key for every value.

2. Related Work

Question answering benchmarks built from financial reports. [FinQA](#) (Chen et al., 2021) built over 8,000 question–answer pairs from earnings reports of S&P 500 companies dated 1999 to 2019. Financial experts wrote the questions and annotated the reasoning steps for each answer, and pre-trained models fell far short of expert performance. [TAT-QA](#) (Zhu et al., 2021) built 16,552 questions over 2,757 passages from real financial reports, each pairing a table with text. Its best model reached 58.0% F1, against 90.8% for human experts. In both benchmarks the model receives the relevant table and text with each question, and answers are scored against labels written by people.

Evaluation of LLMs on SEC filings with retrieval. [FinanceBench](#) (Islam et al., 2023) built 10,231 questions about 40 U.S. public companies from about 360 filings, most of them 10-Ks. The authors tested 16 model configurations on a 150-case sample and reviewed the 2,400 answers by hand. The retrieval setup drove the results. GPT-4-Turbo answered 19% of cases correctly with one vector store shared across all filings, 50% with one store per filing, and 85% when given the evidence pages directly. This study finds the same pattern

for extraction from 10-K statements. It adds an answer key from each company's XBRL data, which grades every value automatically, and an outcome measured at the level of the credit decision.

Accounting-based bankruptcy scores. Altman (1968) introduced the Z-score, a discriminant model of five financial ratios fitted on publicly traded manufacturers, one of which uses the market value of equity. Altman, Hartzell and Peck (1995) introduced the Z'' variant for manufacturers and non-manufacturers, first applied to emerging-market firms. It drops the sales-to-assets ratio, which Altman found sensitive to industry differences, and uses the book value of equity, so it also applies to private firms. In a fifty-year review, [Altman \(2018\)](#) describes this family of models and cautions against relying on the original cutoff scores, because the distribution of corporate credit quality has shifted since the 1960s. Ohlson (1980) proposed a probabilistic bankruptcy model based on financial ratios, and Zmijewski (1984) examined methodological issues in estimating financial distress prediction models. [Grice and Dugan \(2003\)](#) re-estimated the Ohlson and Zmijewski models on 1985–1987 data and tested them on 1988–1991 firms. The coefficients changed markedly. On the hold-out sample, Ohlson's original model classified 39.8% of firms correctly, against 88.7% after re-estimation, and the authors recommend refitting both models on recent data. They also report that re-estimating Altman's model on samples close to the test period improved its accuracy (Grice and Ingram, 2001).

Supervisory guidance on model risk. The Federal Reserve and the Office of the Comptroller of the Currency issued [SR 11-7](#) (2011), supervisory guidance on model risk management. It names three core elements of validation: evaluation of conceptual soundness, ongoing monitoring, and outcomes analysis. This report follows that structure.

3. Data

Extraction sample. Firms were drawn from SEC bulk XBRL data for fiscal year 2024 and kept if they (1) file Form 10-K, (2) tag all six Z'' inputs, (3) hold at least \$50M in total assets, and (4) belong to retail (SIC 5200–5999, which includes restaurants) or software and IT services (SIC 7370–7379). Up to 60 firms per Z'' zone were sampled with a fixed random seed (2026), giving 180 firms. Of these, 153 were kept. A firm was kept if its FY2024 10-K could be retrieved and its statements print current assets, current liabilities, and operating income. Of the 27 firms excluded, 18 had no operating income line in the statement text or the XBRL tags, and 9 failed at the download step. The 153 kept firms comprise 61 retail and 92 software firms, with 49 in the distress zone, 53 in the grey zone, and 51 in the safe zone.

Answer key. The answer key is each firm's own XBRL values from the same 10-K, matched by filing accession number. Total equity uses the tag for equity including noncontrolling interests, or stockholders' equity when that tag is absent. Total liabilities is computed as total assets minus total equity, since not every balance sheet prints a total liabilities line.

Model input. An excerpt of about 60,000 characters holding the balance sheet and income statement, located by heading search.

Backtest population. All 10-K filers with every Z" input and at least \$50M in assets, excluding financial firms, in two cohorts:

Cohort	Values from	Bankruptcies counted in	Firms	Bankruptcies
FY2023	FY2023 filings	2024	2,712	32
FY2024	FY2024 filings	2025	2,612	14

A bankruptcy is an 8-K reporting Item 1.03 (Bankruptcy or Receivership). Firms that reported a bankruptcy before the follow-up year were dropped.

4. Conceptual Soundness

Each row below names an assumption the system relies on, the conditions under which it fails, and the evidence for that failure in this study.

Assumption	Where it fails	Evidence in this study
The excerpt holds the correct statements	Headings split mid-word; titles without "Consolidated"	In run 1, 54 of 56 missing values were never in the excerpt
Each field maps to one printed line	Redeemable noncontrolling interests reported outside equity	For INOD, CCC, and PLBY, the models summed two lines as the prompt asked
The model reads the stated unit	"(In thousands)" split across lines or far from the table	Rule-based challenger before its parsing fix; neither LLM
Operating income approximates EBIT	Firms print no operating income line	18 firms excluded by the inclusion rule

Assumption	Where it fails	Evidence in this study
Z'' weights suit current firms	Retained earnings can misstate current health, both for firms with large accumulated deficits and for firms with large retained earnings	Backtest refit gives retained earnings a small positive coefficient; WW International
The zone cutoffs fit current firms	The 1.10 and 2.60 cutoffs apply to Z'' without its constant term; Altman (2018) presents Z'' with a constant of 3.25 and maps scores to bond-rating equivalents	A constant shifts every score equally, so the AUC results are unaffected; the zone results depend on the cutoffs. The cutoffs used here are the ones commonly applied to Z'' without its constant; no primary source for them was confirmed for this report.

5. Outcomes Analysis: Extraction

This section reports how accurately each method extracted the six inputs, how its errors arose, and how those errors changed each firm's risk zone.

Unless stated otherwise, results use the main prompt at temperature 0. A value counts as correct only if it matches the filed value exactly.

Method	Field accuracy	Wrong values	Missing values	Firms not in correct zone	Average cost per call
GPT-6 Luna, run 1 (before fix)	93.6%	1	56	12 of 148	\$0.0007
GPT-6 Luna	99.9%	1	0	0 of 153	\$0.0007
Claude Haiku 4.5	100%	0	0	0 of 153	\$0.018
Rule-based challenger	89.7%	57	38	32 of 153	none

A firm counts as not in its correct zone if its zone differs from the zone computed with filed values, or if a missing value leaves it unscored. All 12 such firms in run 1 were unscored.

Cost per call is total OpenRouter spend divided by saved calls: \$1.79 over 2,678 calls for GPT-6 Luna and \$14.43 over 824 calls for Claude Haiku. Both averages include calls made on excerpts that were later replaced.

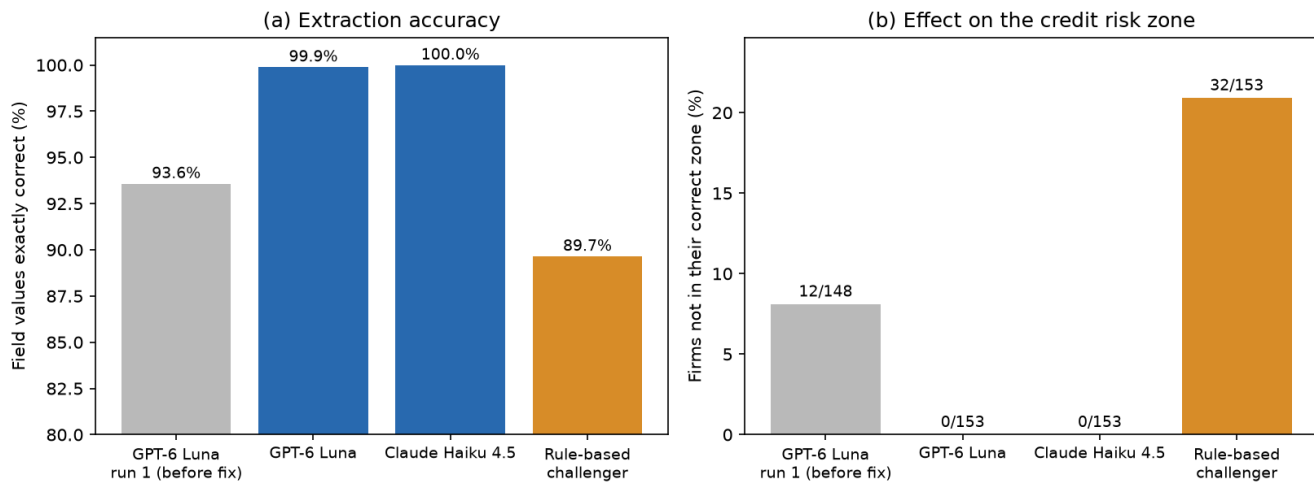


Figure 1. Extraction methods compared on the main prompt. (a) Share of field values matching the filed value exactly; the y-axis starts at 80%. (b) Share of firms not placed in their correct Z" zone, either placed in another zone or left unscored by a missing value. Run 1 is GPT-6 Luna before the statement-search fix, on 148 firms.

Retrieval. In run 1, 54 of the 56 missing values were never in the excerpt the model received, so the model had no number to read. Two causes were confirmed. At AEO and GCO, a heading was split mid-word in the filing text, with "Consolidated B" and "alance Sheets" on separate lines. At MSFT, the balance sheet is titled "Balance Sheets," without "Consolidated." The other affected firms were not diagnosed one by one. The revised search ignores spaces and line breaks and treats "Consolidated" as optional. After the fix, field accuracy rose from 93.6% to 99.9%, unscored firms fell from 12 to 0, and 5 more firms passed the inclusion rule.

Values not printed in the filing. No model invented a number. Every value absent from the text was the sum of two printed lines:

Firm	Printed stockholders' equity	Redeemable noncontrolling interest	Value returned
INOD	63,472	(83)	63,389
CCC	1,997,823	21,679	2,019,502
PLBY	(7,731)	(208)	(7,939)

Amounts in thousands of dollars. All three firms report a redeemable noncontrolling interest outside stockholders' equity, between liabilities and equity. The prompt asked for total equity including noncontrolling interests, and no printed line matched that definition, so the models added the two printed lines. The answer key holds the printed stockholders' equity, because these firms tag no equity total that includes the interest. The error lies in the field definition.

Stability. On the 49 firms with repeated runs for both models, GPT-6 Luna varied on 0 of 294 values and Claude Haiku on 2, both total equity. For WMK, one run returned 1,446,030, a value printed elsewhere in the excerpt, against 1,446,031 on the balance sheet. For INOD, the firm in the table above, 3 of Haiku's 7 runs differed from its most common answer. Haiku's repeated runs cover only these 49 firms because of cost, and INOD has 7 runs in place of 10.

Prompt sensitivity. Under the terse prompt, which drops the "(accumulated deficit)" hint, GPT-6 Luna returned no retained earnings value for 7 of the 75 firms with an accumulated deficit, 6 of them in the grey or distress zone. Claude Haiku returned the correct value for all 75.

Rule-based challenger. The challenger was fixed once for general parsing problems (invisible characters, unit notes split across lines, and tables that list the older year first) and then frozen, so it was not tuned to individual firms. Its weakest fields were operating income (79%) and total equity (82%), both items printed under several different labels. Of the 32 firms not in their correct zone, 31 were unscored because of missing values and 1 was placed in another zone. Of its 57 wrong values, 45 fell in 22 firms that were still scored. These firms received a distorted Z'', and all but one stayed in their correct zone. Missing values are visible and can be flagged; a wrong value is not.

6. Outcomes Analysis: Backtest

This section tests whether Z'' ranks firms by their risk of bankruptcy in the following year, first with Altman's weights and then with weights refit on earlier data.

Bankruptcy rate by zone.

Zone	FY2023 cohort	FY2024 cohort
Safe	0 of 1,142 (0%)	2 of 1,075 (0.19%)
Grey	6 of 431 (1.39%)	0 of 443 (0%)

Zone	FY2023 cohort	FY2024 cohort
Distress	26 of 1,139 (2.28%)	12 of 1,094 (1.10%)

Ranking accuracy. AUC is defined in Section 1. Intervals are 95% bootstrap intervals from 1,000 resamples.

Cohort	Model	AUC	95% interval
FY2023	Altman weights	0.819	0.751–0.879
FY2024	Altman weights	0.759	0.596–0.893
FY2024	Weights refit on FY2023	0.907	0.811–0.963

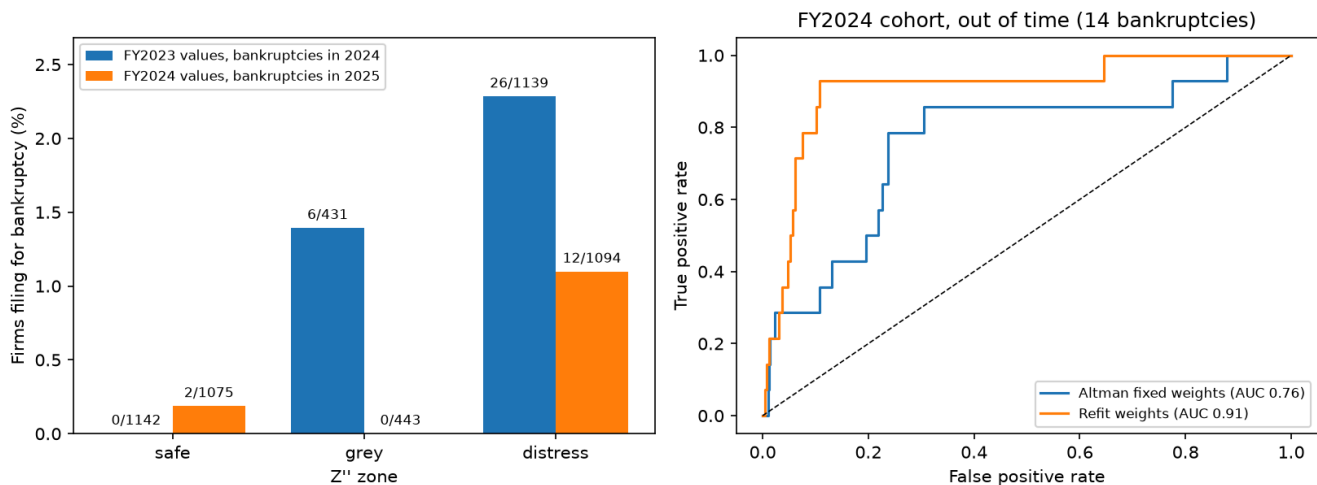


Figure 2. Z'' backtest against SEC bankruptcy filings. (a) Share of firms filing for bankruptcy in the year after their fiscal-year values, by zone; labels show bankruptcies / firms. (b) ROC curves on the FY2024 cohort for Altman's weights and for weights refit on FY2023.

Out-of-time refit. Logistic regression on the four ratios, trained on the FY2023 cohort and tested on FY2024. A paired bootstrap puts the AUC gain at 0.148 (95% interval 0.052–0.283).

Ratio	Altman weight	Refit coefficient
X1 working capital / total assets	6.56	–2.66
X2 retained earnings / total assets	3.26	0.32

Ratio	Altman weight	Refit coefficient
X3 operating income / total assets	6.72	-3.93
X4 total equity / total liabilities	1.05	-0.58

The refit predicts bankruptcy, so a negative coefficient lowers risk. Ratios were capped at the FY2023 1st and 99th percentiles. Compare signs and order across columns, not sizes.

Firms in the safe zone that filed for bankruptcy. WW International had a Z'' of 7.0, well inside the safe zone. Its retained earnings equaled 3.5 times its total assets, while its equity was negative, its working capital was negative, and its operating loss equaled 43% of total assets. The retained earnings term kept the score high while the other three ratios pointed to distress. Zynex had a Z'' of 4.7, with healthy values on all four ratios. It filed for Chapter 11 on December 15, 2025, and [case summaries](#) attribute the filing to a liquidity crisis after TRICARE, a government health insurer, suspended payments in March 2025 pending an investigation of billing practices. That event left no trace in the prior year's ratios.

7. Limitations

- **Sample scope.** The extraction sample covers two sectors and one fiscal year, and it includes only firms whose statements print every input. Results may differ for other industries, older filings, or statements with less standard layouts.
- **Answer key.** The answer key is each company's own XBRL tagging, which can differ from the printed statement. For the three firms with redeemable noncontrolling interests, the key matched the printed stockholders' equity line.
- **Stability coverage.** Claude Haiku's repeated runs cover 49 firms because of cost, so its stability rate is less certain than GPT-6 Luna's, which covers all 153.
- **Backtest events.** The cohorts include 32 and 14 bankruptcies, which leaves wide intervals around each AUC. The test spans one out-of-time year and a one-year horizon, and bankruptcies not reported in an 8-K are missed.
- **Model versions.** Results hold for the model versions tested at the time of this study. A new model version requires a new validation.

8. Recommendations and Controls

Each control below targets a failure observed in this study.

1. **Run extraction at temperature 0.** Every case of run-to-run variation (WMK, INOD, CCC, PLBY, PPLI) appeared in repeated runs at the default temperature.
2. **Check that each returned value appears as a printed number in the excerpt.** In this study, this check flagged every total equity value built by adding lines (INOD, CCC, PLBY).
3. **Flag any missing value for human review.** All 12 unscored firms in run 1 and 31 of the challenger's 32 firms out of their correct zone traced to missing values, which this flag would catch before scoring.
4. **Verify that the excerpt contains the statements before extraction,** for example by requiring "Total assets" and an income line. This check targets the retrieval failures of run 1, in which the excerpt lacked the statement the model needed.
5. **Define each field to match how filings present it,** return the printed line, and never combine lines. This removes the field-definition error behind INOD, CCC, and PLBY.
6. **Keep the label hint for retained earnings ("accumulated deficit") in the prompt.** Without it, GPT-6 Luna missed 7 firms, 6 of them in the grey or distress zone.

With these controls in place, GPT-6 Luna is the better choice for production. It cost about 26 times less per call than Claude Haiku, its only main-prompt error came from the field definition that control 5 fixes, and its misses on accumulated deficits occurred only without the hint that control 6 keeps. It also varied less across repeated runs on the 49 firms tested with both models. Claude Haiku suits a setting where these controls cannot be added, at a higher cost.

9. Ongoing Monitoring

- **Accuracy.** Each quarter, score a sample of new extractions against XBRL, where it exists, and against hand-checked values for statements without it.
- **Zone changes.** Each quarter, report the share of firms whose zone changes when extracted values replace checked values.
- **Retrieval.** Each month, track the share of excerpts that fail the statement check in control 4. A rise signals a new filing layout.
- **Model versions.** Repeat the full validation after any change to the model version or the prompt.
- **Z" weights.** Re-estimate the weights each year as new bankruptcies are observed, and compare them with Altman's weights out of time, as in Section 6.